Smart Working Station

Project idea, description and objectives

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1. Idea

Hands-on desk work, such as electronics, assembly or design, happens on a physical surface, while the digital tools it needs live on a separate screen, keyboard and mouse. Every look-up breaks the flow of work.

The Smart Working Station puts the digital interface on the work surface itself. A projector displays controls and information directly on the desk. A transparent touch foil turns that projected area into a touchscreen. A voice assistant lets the user ask questions and trigger actions without taking their hands off the work.

2. Description

The system is built around a Raspberry Pi 5 and combines four interaction channels:

- **Projected interface:** a projector above the desk shows a user interface (currently an assistant status panel) on a flat work surface, sized for about a 32-inch image.
- **Touch input:** a 32-inch PCAP capacitive touch foil with an ILITEK 2510 USB controller makes the projected area touch-sensitive.
- **Voice interaction:** a USB microphone and an on-device wake word ("Hey Atlas"), or a push-to-talk button, start a conversation. Speech is transcribed, answered by a language model that can read project files through restricted read-only tools, and spoken back. The Atlas model is selected for noncommercial personal and Academy demonstrations.
- **Physical controls and audio:** a custom Raspberry Pi 5 HAT (add-on circuit board) carries two MAX98357A I2S amplifiers for stereo speech output, plus connections for an AI/push-to-talk button with LED, a volume/mute rotary encoder, a restart button and a power button.

The project covers hardware (PCB design, audio electronics, projector and touch integration), embedded Linux software in Python, and mechanical integration of the workstation.

3. Objectives

1. Project a clear, readable interface onto a flat work surface, with measured geometry (throw distance, image size, full-frame display).
2. Make the projected surface touch-sensitive and map touch positions onto the projected interface.
3. Provide a hands-free voice assistant with a wake word and a push-to-talk button, with short response times.
4. Keep AI actions safe: the assistant may only call an allowlisted set of actions (currently read-only file tools), never unrestricted system commands, and destructive actions require confirmation.
5. Design a custom Raspberry Pi 5 HAT PCB that integrates stereo audio amplification and the physical controls.
6. Keep the system modular, testable subsystem by subsystem, and maintainable.

4. System architecture

Subsystem	Main components
Processing	Raspberry Pi 5 (4 GB), 32 GB microSD, active cooler
Display	CAIWEI A12AB LED projector, native 1920×1080, connected over HDMI
Touch	32-inch PCAP touch foil with ILITEK 2510 USB controller
Voice input	USB microphone; openWakeWord on the Pi; webrtcvad end-of-speech detection
Voice processing	Cloud speech-to-text → language model with read-only project tools → cloud text-to-speech (streamed)
Audio output	Current setup: the projector's built-in speakers over HDMI. Designed: 2 × MAX98357AETE+T I2S Class-D amplifiers on the HAT driving two small speakers (left and right). The final choice between them is still open.
Controls	AI/push-to-talk button with LED, rotary encoder (volume + mute), restart button (2-second hold), power button wired to the Pi's power pads
Software	Python on Raspberry Pi OS; assistant, UI server and restart handler run as separate processes, so one failing cannot take down the others

5. Current progress

Item	Status
Voice assistant (wake word / button → answer spoken back)	Pipeline verified on the Raspberry Pi with the previous Hey Jarvis trigger; about 3.6 s from end of speech to first sound in that earlier test. A complete Hey Atlas conversation was heard through the projector on 2026-09-29; recognition reliability remains under evaluation.
Projected status panel	Working and verified on the projector
Projector (CAIWEI A12AB)	Native 1920×1080 confirmed; throw ratio measured at 1.28–1.36 (a 32-inch image needs about 0.91–0.96 m); edge cropping (overscan) fixed. Focus at that short distance not yet tested
Restart button	Working and verified on the Raspberry Pi
Audio output	A complete spoken reply was heard through the projector speakers over HDMI (2026-09-29). Replies were later reported cutting out during a longer test; the cause is being investigated. One MAX98357A amplifier channel (bench module) works; stereo is pending a replacement module
Custom HAT PCB	Schematic, layout, BOM and Gerber files exported from EasyEDA; under instructor review; not yet fabricated
Touch foil	Not yet tested
Projector mount / mechanical design	Research done: a load-rated four-arm projector mount and a braced stand are being evaluated; the projector's mounting holes, weight and lens-down use still need to be measured and approved. No 3D files yet

6. Next steps

1. Test projector focus at the desk distance (~0.91–0.96 m); this decides the mounting geometry.
2. Find the cause of the reply cutouts on the projector speakers, then retest the push-to-talk button.
3. Test the touch foil, then map touch positions onto the projected interface.
4. Decide the final audio output (projector speakers or HAT amplifiers) after a loudness comparison.
5. Resolve the instructor's PCB review and the amplifier power input, then fabricate and test the HAT.
6. Start the assistant, UI and restart button automatically at boot.
7. Measure the projector mounting pattern, then design and get approval for the mount and stand (3D).

7. Scope notes

An earlier version of the project used a camera and computer vision for object recognition. That direction was removed, so the project now focuses on projected touch and voice interaction. A voice-driven parametric CAD → 3D-printing feature is being considered as a future extension only.

Status words follow the project convention: "designed" means drawn, not built; "tested" means a real test was run; "verified" means recorded evidence supports the result.